Alexandria University
Faculty of Engineering
Computer and Systems Engineering Dept.
CSE232 : Digital logic design

Lab #3 Report

Sequence Detector

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1 Problem statement

The objective of this project is to design and implement a synchronous sequential circuit using VHDL. The circuit is required to process a serial input stream and generate a specific output based on the history of the inputs received.

The circuit must have one input (X) and one Moore-type output (Z). The output (Z) should be asserted ($Z = 1$) if and only if the following two conditions are met simultaneously:

- **Condition 1**: The total number of 1's received since the start of operation is divisible by 4 (i.e., $count(1) \pmod 4 = 0$).
- **Condition 2**: The total number of 0's received since the start of operation is an odd number (i.e., $count(0)$ is odd).

Additionally, the design will be extended to follow the Mealy model, where the output (Z) depends on both the current state and the current input (X). A comparison between the Moore and Mealy implementations will be conducted to analyze differences in state count, timing, and output behavior for the same input sequence.

2 Design and Analysis

2.1 Moore machine

2.1.1 State diagram

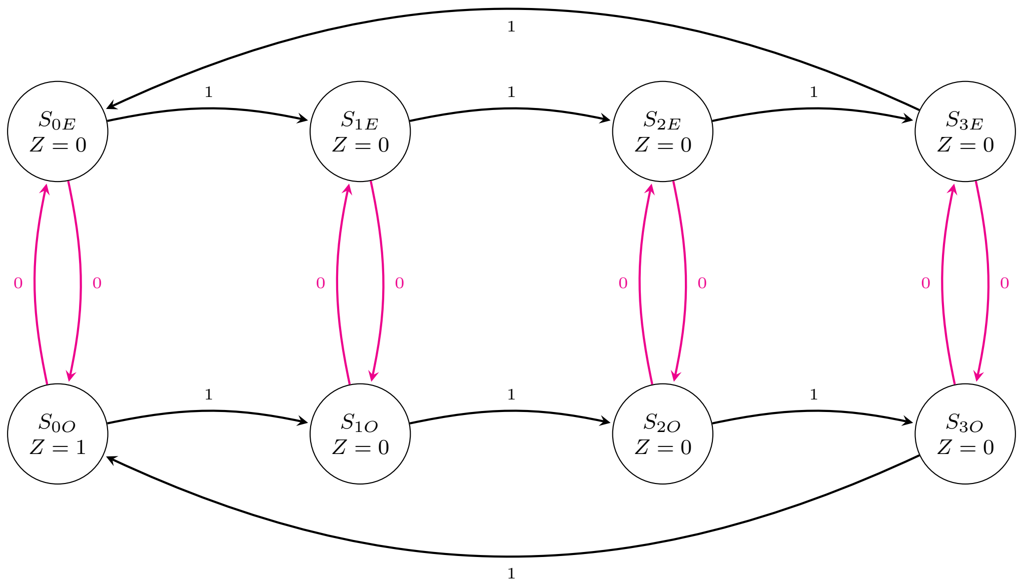


Figure 1: State Transition Diagram for the Moore Machine

Legend of State Diagram:
S_{ij} such that:

- i = Number of 1's mod 4.
- $j = \begin{cases} E & \text{Number of 0's is even} \\ O & \text{Number of 0's is odd} \end{cases}$

2.1.2 State table

Current State (S_{ij})	**Next State** (Q^+)		**Output** (Z)
	$X = 0$	$X = 1$	
S_{0E} (0)	S_{0O} (4)	S_{1E} (1)	0
S_{1E} (1)	S_{1O} (5)	S_{2E} (2)	0
S_{2E} (2)	S_{2O} (6)	S_{3E} (3)	0
S_{3E} (3)	S_{3O} (7)	S_{0E} (0)	0
S_{0O} (4)	S_{0E} (0)	S_{1O} (5)	1
S_{1O} (5)	S_{1E} (1)	S_{2O} (6)	0
S_{2O} (6)	S_{2E} (2)	S_{3O} (7)	0
S_{3O} (7)	S_{3E} (3)	S_{0O} (4)	0

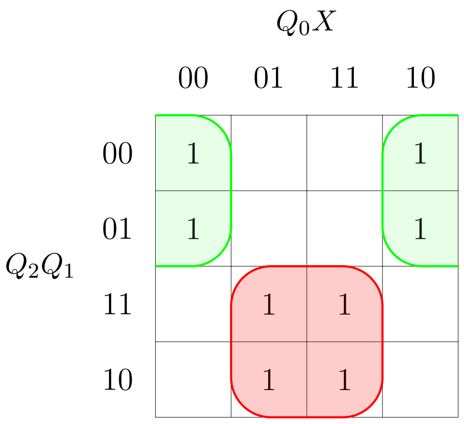
Table 1: State Table for the Moore Sequential Circuit

2.1.3 Transition table

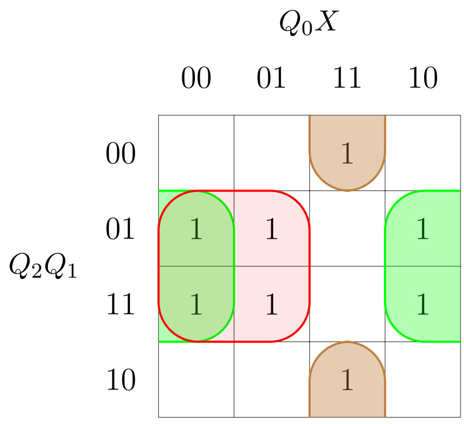
Current State			**Next State** (Q^+)						**Output (Z)**
			$X = 0$			$X = 1$			
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	Q_2^+	Q_1^+	Q_0^+	
0	0	0	1	0	0	0	0	1	0
0	0	1	1	0	1	0	1	0	0
0	1	0	1	1	0	0	1	1	0
0	1	1	1	1	1	0	0	0	0
1	0	0	0	0	0	1	0	1	1
1	0	1	0	0	1	1	1	0	0
1	1	0	0	1	0	1	1	1	0
1	1	1	0	1	1	1	0	0	0

Table 2: Transition Table with Binary Assignments for Moore Machine

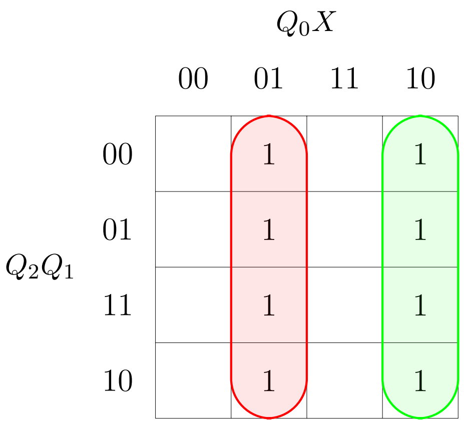
2.1.4 Karnaugh maps



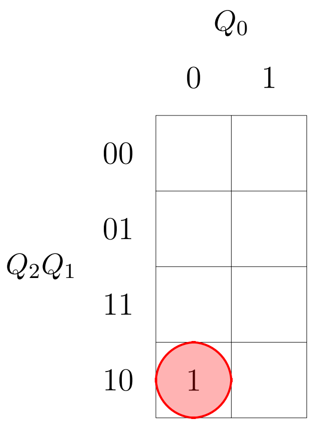
(a) K-map for Flip-flop Q_2^+



(b) K-map for Flip-flop Q_1^+



(c) K-map for Flip-flop Q_0^+



(d) K-map for output Z

Figure 2: K-maps for Moore Machine Flip-flops

2.1.5 Equations

$$D_2 = Q_2^+ = Q_2 \cdot X + \overline{Q_2} \cdot \overline{X} = Q_2 \odot X \tag{1}$$

$$D_1 = Q_1^+ = Q_1 \cdot \overline{Q_0} + Q_1 \cdot \overline{X} + \overline{Q_1} \cdot Q_0 \cdot X \tag{2}$$

$$D_0 = Q_0^+ = \overline{Q_0} \cdot X + Q_0 \cdot \overline{X} = X \oplus Q_0 \tag{3}$$

$$Z = Q_2 \cdot \overline{Q_1} \cdot \overline{Q_0} \tag{4}$$

2.1.6 Important assumptions

- The circuit is implemented using **D Flip-flops** for each of the three bits (D_2, D_1, D_0).

$$D_n = Q_n^+$$

- **Flip-flops:** All flip-flops are assumed to be **positive-edge** triggered.
- **Propagation Delay:** The propagation delay of the logic gates is **neglected** in the theoretical analysis.
- The input X is assumed to be **synchronous** and stable during the setup and hold time intervals around the rising edge of the clock. (Changes only on the falling edge of the clock.)
- **Clock Signal:** The clock signal is assumed to be a regular square wave with a constant frequency, and it is assumed to reach all flip-flops simultaneously.
- **Clock Period:** It's assumed to be 10 ns.
- **Reset State:** The reset signal is **asynchronous**, meaning the circuit is forced to zero if **Reset** is equal to one, **independent** of the clock edge and **regardless of** the value of the input X.

2.1.7 Simulation samples

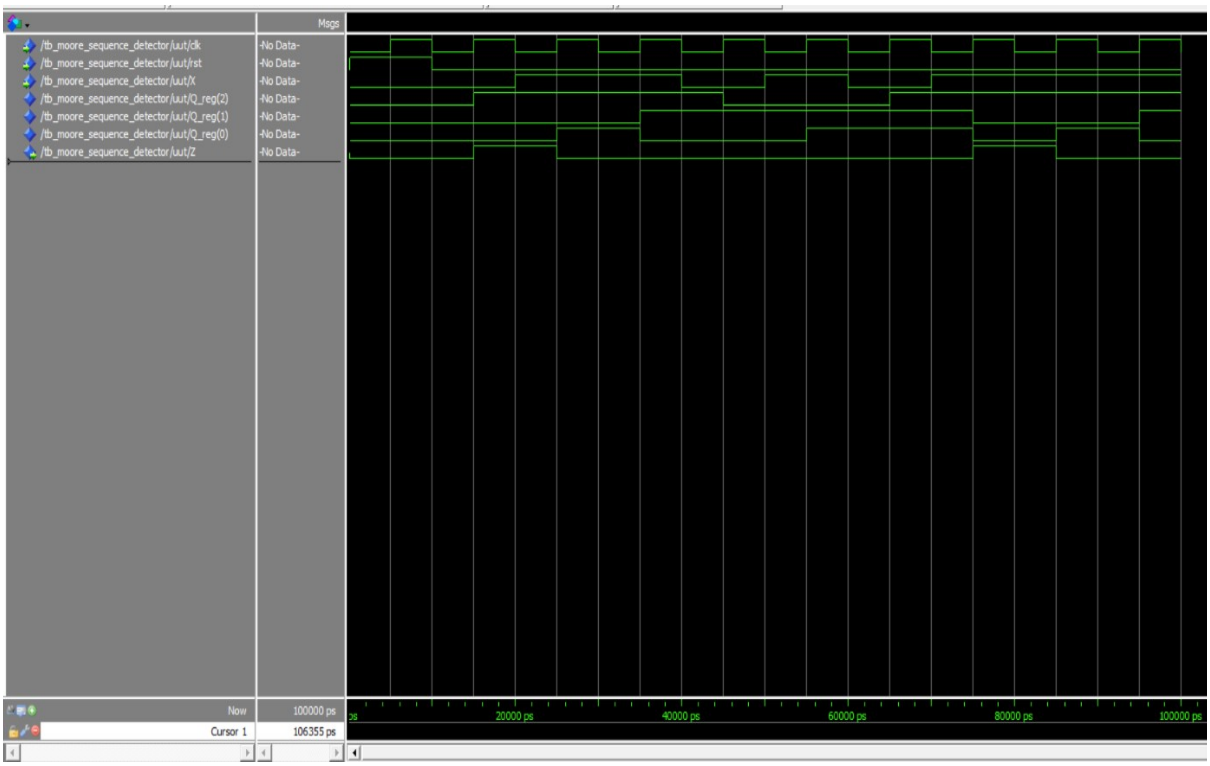


Figure 3: Simulation Sample for Moore Machine.

2.2 Mealy machine

2.2.1 State diagram

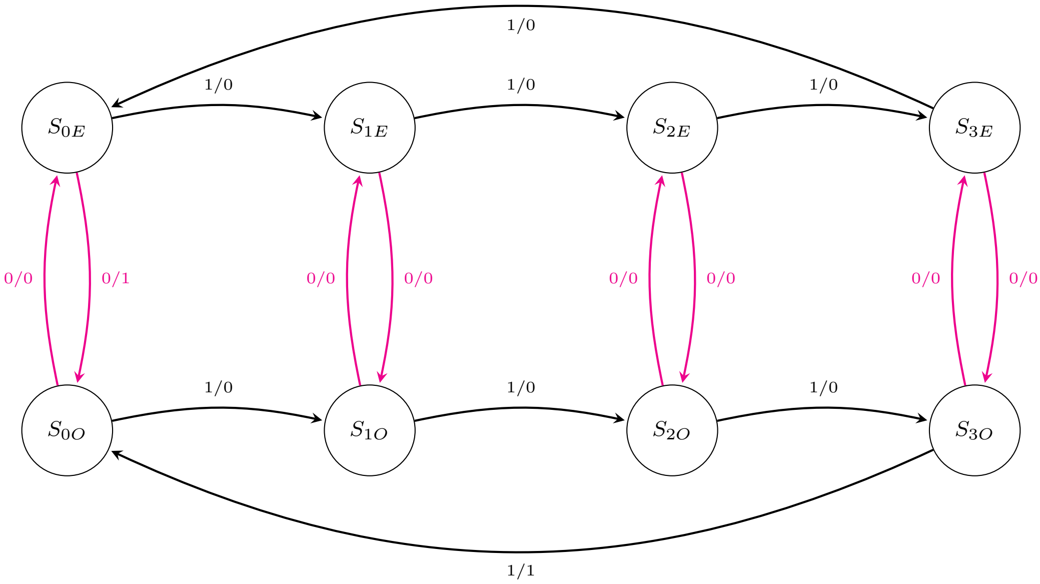


Figure 4: State Transition Diagram for the Mealy Machine

Legend of State Diagram:
S_{ij} such that:

- i = Number of 1's mod 4.
- $j = \begin{cases} E & \text{Number of 0's is even} \\ O & \text{Number of 0's is odd} \end{cases}$

2.2.2 State table

Current State (S_{ij})	**Next State** (Q^+) $X=0$	$X=1$	**Output (Z)** $X=0$	$X=1$
S_{0E} (0)	S_{0O} (4)	S_{1E} (1)	1	0
S_{1E} (1)	S_{1O} (5)	S_{2E} (2)	0	0
S_{2E} (2)	S_{2O} (6)	S_{3E} (3)	0	0
S_{3E} (3)	S_{3O} (7)	S_{0E} (0)	0	0
S_{0O} (4)	S_{0E} (0)	S_{1O} (5)	0	0
S_{1O} (5)	S_{1E} (1)	S_{2O} (6)	0	0
S_{2O} (6)	S_{2E} (2)	S_{3O} (7)	0	0
S_{3O} (7)	S_{3E} (3)	S_{0O} (4)	0	1

Table 3: State Table for the Mealy Sequential Machine

2.2.3 Transition table

Current State			**Next State** (Q^+)						**Output (Z)**	
			$X=0$			$X=1$				
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	Q_2^+	Q_1^+	Q_0^+	$X=0$	$X=1$
0	0	0	1	0	0	0	0	1	1	0
0	0	1	1	0	1	0	1	0	0	0
0	1	0	1	1	0	0	1	1	0	0
0	1	1	1	1	1	0	0	0	0	0
1	0	0	0	0	0	1	0	1	0	0
1	0	1	0	0	1	1	1	0	0	0
1	1	0	0	1	0	1	1	1	0	0
1	1	1	0	1	1	1	0	0	0	1

Table 4: Transition Table with Binary Assignments for Mealy Machine

2.2.4 Karnaugh maps

Q_2Q_1 \ Q_0X	00	01	11	10
00	1			1
01	1			1
11		1	1	
10		1	1	

(a) K-map for Flip-flop Q_2^+

Q_2Q_1 \ Q_0X	00	01	11	10
00			1	
01	1	1		1
11	1	1		1
10			1	

(b) K-map for Flip-flop Q_1^+

Q_2Q_1 \ Q_0X	00	01	11	10
00		1		1
01		1		1
11		1		1
10		1		1

(c) K-map for Flip-flop Q_0^+

Q_2Q_1 \ Q_0X	00	01	11	10
00	1			
01				
11			1	
10				

(d) K-map for output Z

Figure 5: K-maps for Mealy Machine Flip-flops

2.2.5 Equations

$$D_2 = Q_2^+ = Q_2 \cdot X + \overline{Q_2} \cdot \overline{X} = Q_2 \odot X \tag{5}$$

$$D_1 = Q_1^+ = Q_1 \cdot \overline{Q_0} + Q_1 \cdot \overline{X} + \overline{Q_1} \cdot Q_0 \cdot X \tag{6}$$

$$D_0 = Q_0^+ = \overline{Q_0} \cdot X + Q_0 \cdot \overline{X} = X \oplus Q_0 \tag{7}$$

$$Z = \overline{Q_2} \cdot \overline{Q_1} \cdot \overline{Q_0} \cdot \overline{X} + Q_2 \cdot Q_1 \cdot Q_0 \cdot X \tag{8}$$

2.2.6 Important assumptions

- The counter is implemented using **D Flip-flops** for each of the four bits (D_3, D_2, D_1, D_0).
$$D_n = Q_n^+$$
- **Flip-flops:** All flip-flops are assumed to be **positive-edge** triggered.
- **Propagation Delay:** The propagation delay of the logic gates is **neglected** in the theoretical analysis.
- The input X is assumed to be **synchronous** and stable during the setup and hold time intervals around the rising edge of the clock. (Changes only on the falling edge of the clock.)
- **Clock Signal:** The clock signal is assumed to be a regular square wave with a constant frequency, and it is assumed to reach all flip-flops simultaneously.
- **Clock Period:** It's assumed to be 10 ns.
- **Reset State:** The reset signal is **asynchronous**, meaning the counter is forced to zero if **Reset** is equal to one, **independent** of the clock edge and **regardless of** the value of the input X.
- **Asynchronous Output:** If the input X changes, the output Z changes immediately and does not wait for the next rising edge.

2.2.7 Simulation samples

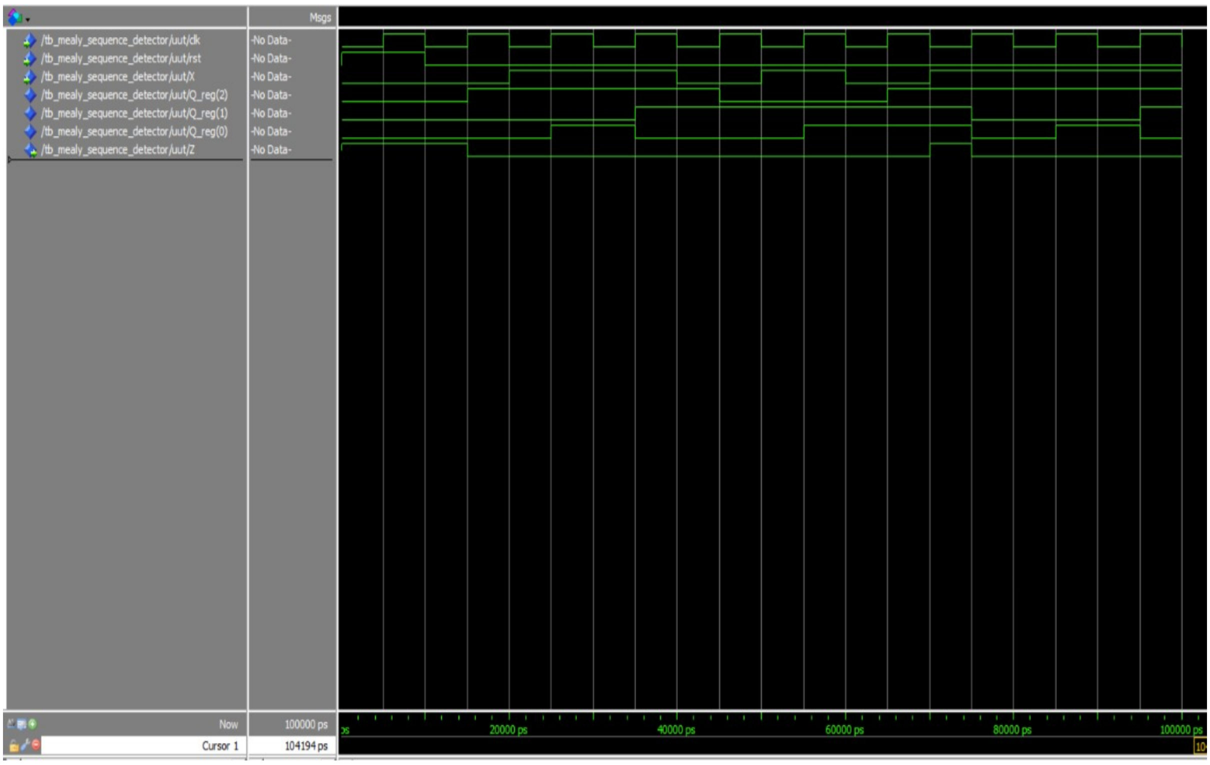


Figure 6: Simulation Sample for Mealy Machine.

3 Comparison Between Moore and Mealy Designs

3.1 Functional Comparison

While both machines successfully solve the sequence detection problem, their structural and timing behaviors differ significantly. Based on the derived K-maps and transition tables, the primary differences are as follows:

1. **State Memory and Transition Logic (Identical)**
   - **Number of States**: Unlike many sequence detectors where a Mealy machine requires fewer states, this specific problem requires the circuit to continuously "remember" two independent facts: the 1's count modulo-4 (4 conditions) and the 0's parity (2 conditions). Because $4 \times 2 = 8$, both the Moore and Mealy designs require exactly 8 states and 3 flip-flops (Q_2, Q_1, Q_0).
   - **Next-State Logic**: Because the state mapping is identical, the Boolean equations for the D flip-flops (D_2, D_1, D_0) are exactly the same for both machines.
2. **Output Logic Complexity (Different)**
   - **Moore Machine**: The output Z relies entirely on the current state. The combinational logic is highly minimized: $Z = Q_2 \cdot \overline{Q_1} \cdot \overline{Q_0}$. This requires only a single 3-input AND gate.
   - **Mealy Machine**: The output Z relies on both the current state and the current input X. The derived logic is more complex: $Z = \overline{Q_2}\cdot\overline{Q_1}\cdot\overline{Q_0}\cdot\overline{X}+Q_2\cdot Q_1\cdot Q_0\cdot X$. This requires more hardware (two 4-input AND gates and one OR gate) to implement.
3. **Timing and Output Reaction (Different)**
   - **Moore Machine (Synchronous)**: The output is synchronized with the clock. It only asserts $Z = 1$ after the active clock edge transitions the circuit into the target state (S_{0O}). This introduces a one-clock-cycle delay relative to the input sequence but guarantees a stable, glitch-free output.
   - **Mealy Machine (Asynchronous Reaction)**: The output reacts immediately to the input X during the clock cycle. If the circuit is in a precursor state (like S_{3O}) and the correct input ($X = 1$) is applied, Z will immediately go high before the clock edge arrives. This makes the Mealy circuit faster to respond but susceptible to "false outputs" (glitches) if the input X fluctuates before stabilizing.

3.2 Output Sequence Snippet (Timing Diagram)

To illustrate the timing difference, here is an output snippet for both circuits based on the exact same input sequence.
Initial Condition: Assume the circuit is reset to S_{0E} (0 ones, even zeros).
Input Sequence (X): 0, 1, 1, 0, 1, 0, 1

Clock Cycle (t)	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Current State	S_{0E}	S_{0O}	S_{1O}	S_{2O}	S_{2E}	S_{3E}	S_{3O}	S_{0O}
Input (X) applied	0	1	1	0	1	0	1	-
Moore Output (Z)	0	**1**	0	0	0	0	0	**1**
Mealy Output (Z)	**1**	0	0	0	0	0	**1**	0

Table 5: Timing Analysis: Moore vs. Mealy Output Reaction

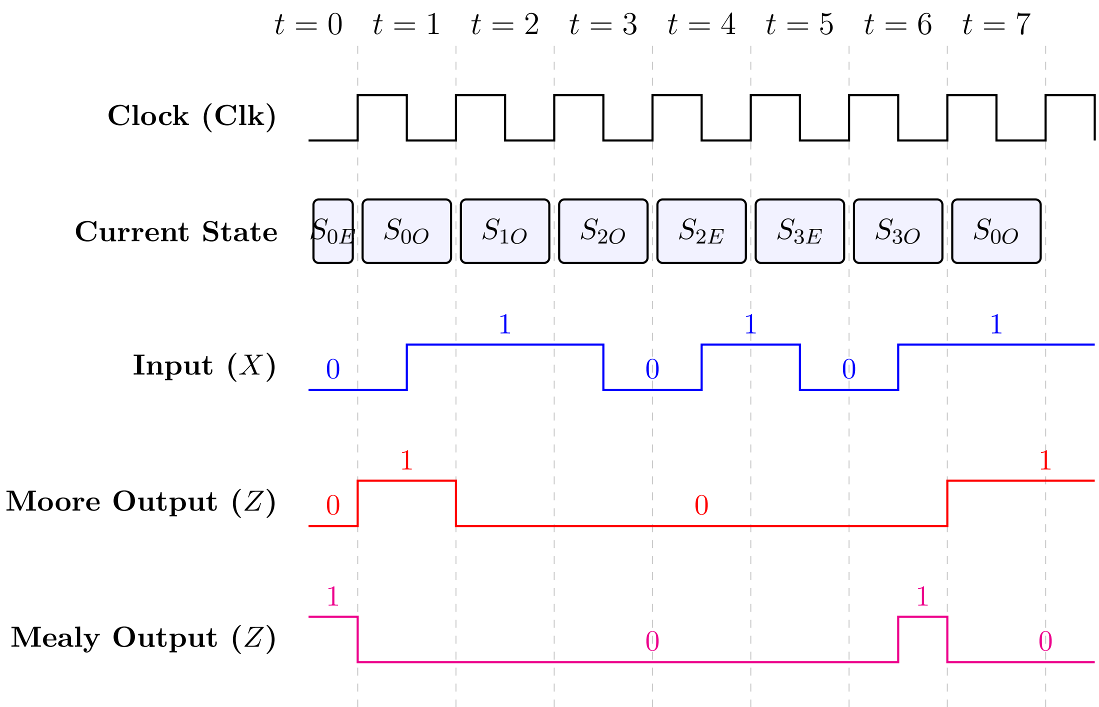


Figure 7: Timing Diagram Illustrating the 1-Cycle Phase Shift between Moore and Mealy Outputs (Input transitions on Falling Edge)